ChunQiuTR: Time-Keyed Temporal Retrieval in Classical Chinese Annals

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Code & Data: github.com/xbdxwyh/ChunQiuTR

Takeaway： Retrieve the right event at the right time.

Motivation

RIGHT-LOOKING EVIDENCE ≠ RIGHT-TIME EVIDENCE

Query

魯莊公二年十二月发生了什么？

What happened in the 12th month of Duke Zhuang's 2nd year?

→ **Target time key**

(Duke Zhuang, Year 2, Month 12)

M−1
adjacent month

M
Duke Zhuang · Year 2 · Month 12

M+1
adjacent month

1 No-event commentary

adjacent key · lexical-near · wrong month

魯莊公二年十一月：
經傳於此月无事可书。

... indicate that nothing was recorded ...

wrong evidence type

2 Broad-season commentary

event-like · chrono-near · misleading evidence

二年冬，夫人姜氏會齊侯於禚。書，奸也。

In winter of Year 2, Lady Jiang met the ...

off-target evidence

3 Canonical record

target month · canonical record · ground truth

冬十有二月，夫人姜氏會齊侯於禚。

... Lady Jiang met the Marquis of Qi at Zhuo.

correct evidence

Top-3 retrieved records

1 [NEG] no-event commentary

2 [NEG] broad-season commentary

3 [RECORD] canonical record

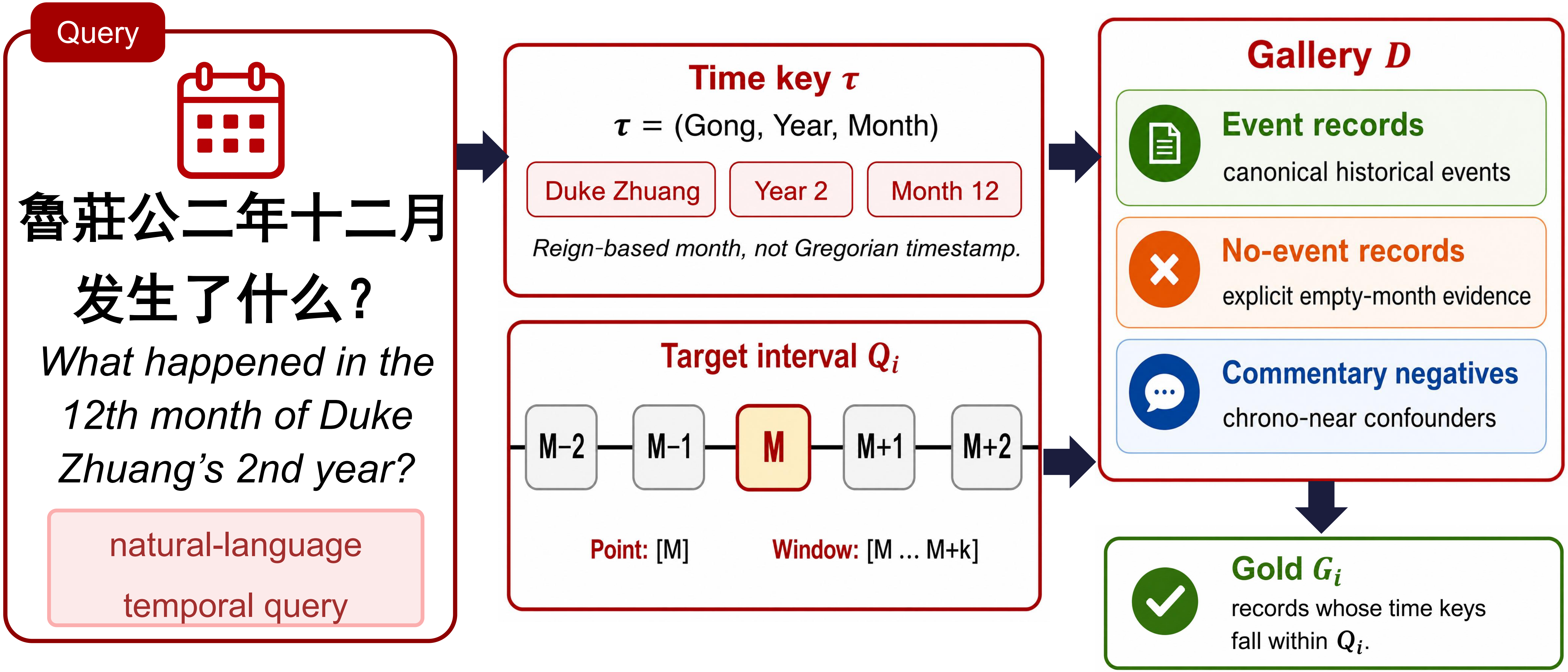
LLM / RAG Answer

Fluent, but grounded in wrong-time evidence.

Right-looking evidence is not enough; it must also be right-time evidence.

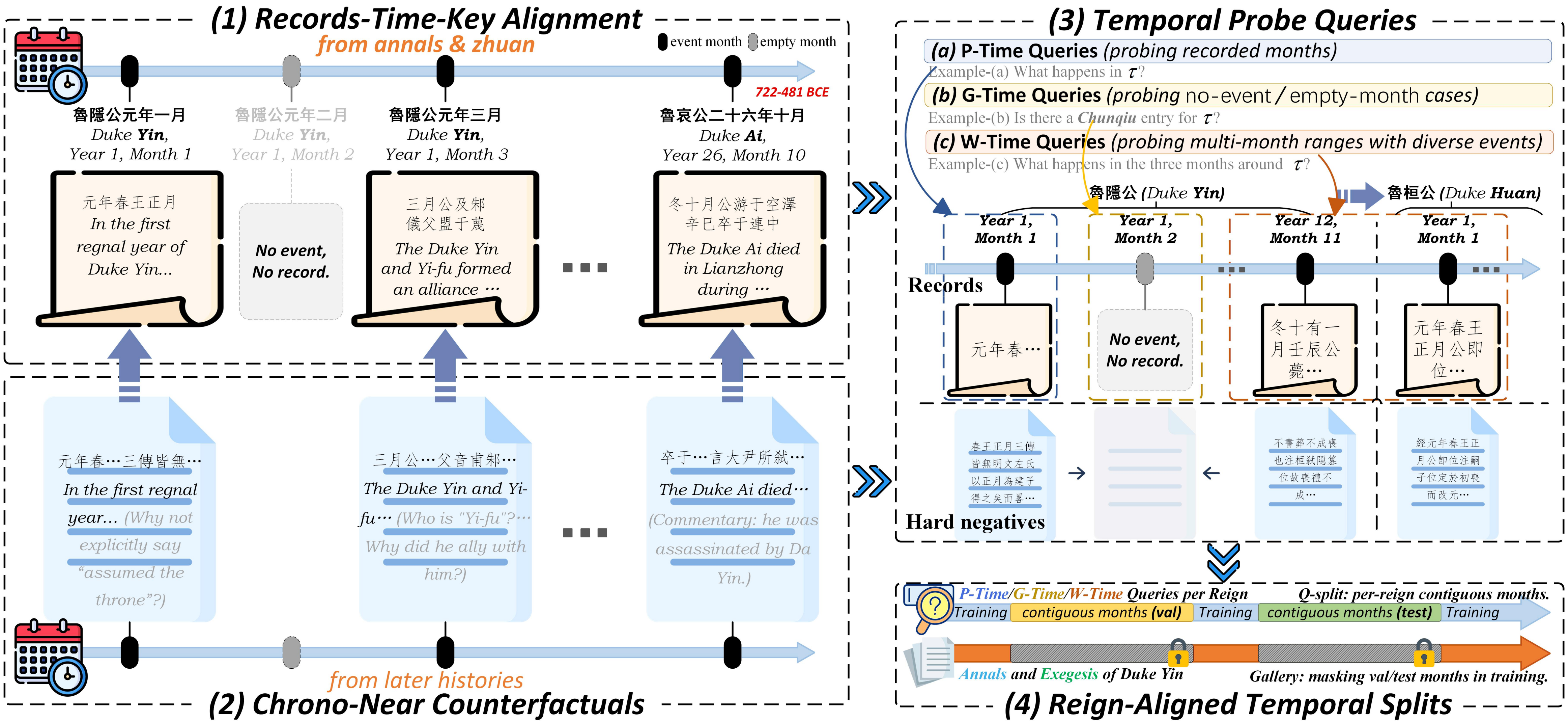
Task

Retrieve Evidence by Reign-based Time Keys



The task tests whether retrieval can respect when, not only what.

ChunQiuTR A Time-keyed Benchmark from Chunqiu and Three Zhuan



ChunQiuTR A Time-keyed Benchmark from Chunqiu and Three Zhuan

Why Chunqiu?

Chunqiu is a compact stress test for time-keyed retrieval:

events are sparse, dates are reign-based, and commentaries create many right-looking but temporally misleading records.

Reign-based time keys

No-event months

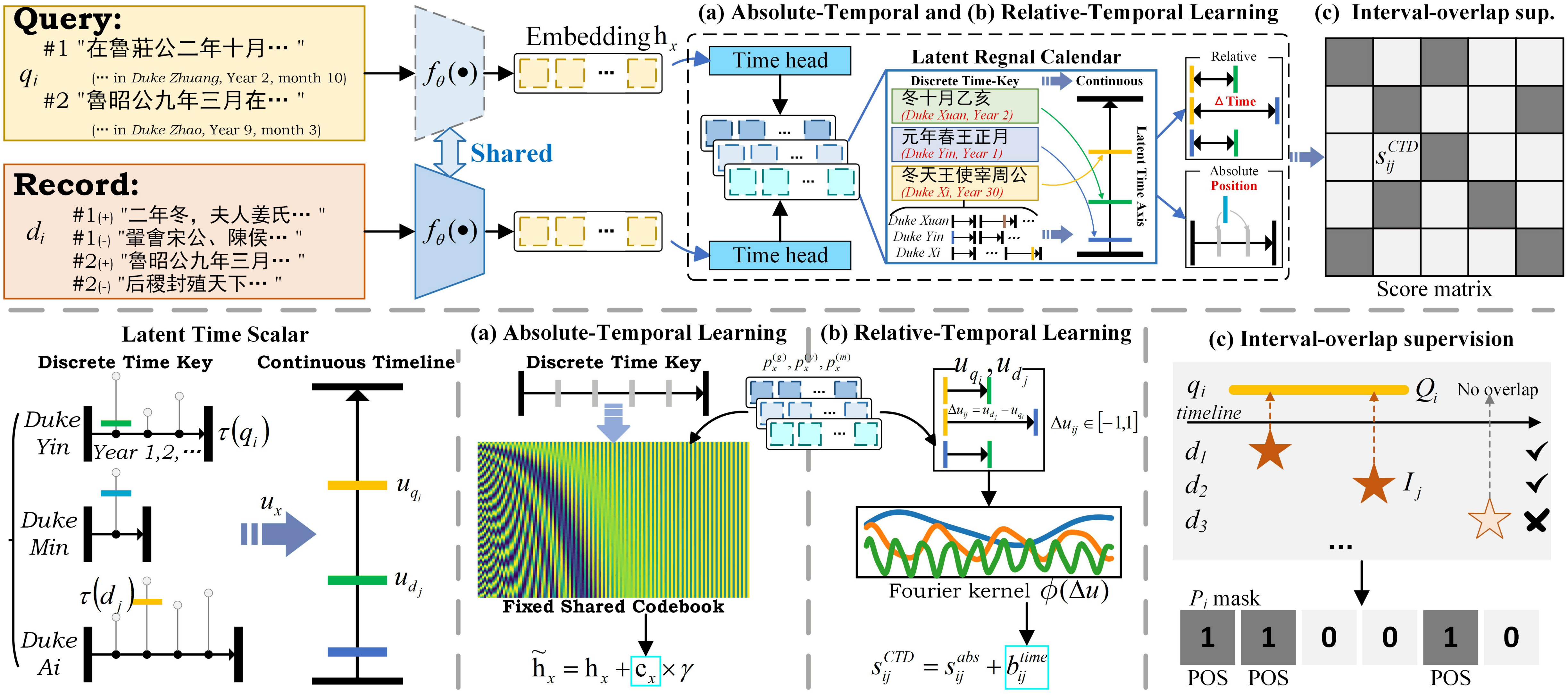
Lexical-near commentaries

Scale & Design

Month keys	**3,036**
Records	**20,172**
Queries	**16,226**
Train / Val / Test	**13,053 / 1,520 / 1,653**
Split	**reign-aligned** (month-level)
Negatives	**chrono-near** (counterfactuals)

Method

CTD: Soft Calendrical Learning, not Hard Metadata Filtering



Method

CTD: Soft Calendrical Learning, not Hard Metadata Filtering

What CTD adds

Semantic relevance + calendrical consistency

Core intuition

- Semantic matching alone is not enough for historical retrieval.
- A retrieved record should be not only topically relevant, but also temporally faithful to the target regnal month.
- CTD augments dual-encoder retrieval with learned calendrical consistency from text.

Three key ingredients

Absolute temporal learning
learns a soft position on the latent regnal calendar

Relative temporal bias
models query–record temporal offsets

Interval-overlap supervision
supports multi-month window retrieval

Challenge → CTD response

Challenge: Reign-based months are not standard timestamps.
→ **Response:** Absolute temporal learning maps text to a latent regnal calendar.

Challenge: Nearby records and commentaries are often plausible but off-target.
→ **Response:** Relative temporal bias suppresses chrono-near confounders.

Challenge: Some queries ask for temporal windows rather than a single month.
→ **Response:** Interval-overlap supervision enables multi-positive window retrieval.

CTD turns topic matching into time-faithful retrieval.

Results

CTD Improves Time-keyed Retrieval

Method	Pub	Type	R@1	R@5	R@10	MRR@10	nDCG@10
Sparse retrieval							
BM25	–	Sparse	0.3962	0.5209	0.5620	0.4487	0.3404
BM25+TimeKDE	–	Sparse	0.4943	0.6086	0.6709	0.5456	0.4222
SPLADE-IDF$_{(ZS)}$	arXiv'24	Sparse	0.1361	0.2765	0.3569	0.1971	0.1596
SPLADE-$\ell_0$$_{(ZS)}$	SIGIR'25	Sparse	0.0006	0.0309	0.0587	0.0132	0.0143
Fusion / late interaction							
ColBERT-JINA$_{(ZS)}$	MRL'24	Fusion	0.2498	0.4102	0.4743	0.3167	0.2569
ColBERT-LFM2$_{(ZS)}$	arXiv'25	Fusion	0.3345	0.4567	0.4894	0.3865	0.2691
Dense retrieval (encoder-based)							
mE5-Large$_{(ZS)}$	arXiv'24	Dense	0.2916	0.3969	0.4574	0.3389	0.2441
mE5-Large-ins$_{(ZS)}$	arXiv'24	Dense	0.2359	0.3545	0.4162	0.2862	0.2358
GTE-Large$_{(ZS)}$	arXiv'23	Dense	0.2293	0.3527	0.3890	0.2826	0.2188
BGE-Large-v1.5$_{(ZS)}$	arXiv'23	Dense	0.2208	0.3430	0.4144	0.2775	0.2280
BGE-m3$_{(ZS)}$	Findings ACL'24	Dense	0.2698	0.3775	0.4253	0.3135	0.2299
BERT-base$_{(FT)}$	NAACL'19	Dense	0.5088	0.6279	0.6727	0.5597	0.4283
BERT-base + TempDate$_{(FT)}$	SIGIR'23	Dense	0.5027	0.6165	0.6691	0.5508	0.4243
BERT-base + TempDate-Smooth$_{(FT)}$	PMLR'23	Dense	0.5051	0.6152	0.6673	0.5519	0.4244
CTD $_{\textbf{BERT-base}}$ (Ours)	This work	Dense	**0.5826**	**0.6721**	**0.7090**	**0.6193**	**0.4575**
Dense retrieval (LM-based embeddings)							
GTE-Qwen2-1.5B$_{(ZS)}$	arXiv'23	Dense	0.2783	0.4453	0.5009	0.3501	0.2613
E5-mistral-7B$_{(ZS)}$	ACL'24	Dense	0.2196	0.3212	0.3684	0.2619	0.2359
PQR (Qwen2.5-7B)$_{(re)}$	ACL'25	Dense	0.1585	0.3134	0.3805	0.2226	0.1712
PQR (Qwen3-8B)$_{(re)}$	ACL'25	Dense	0.0901	0.2184	0.3152	0.1481	0.1184
Qwen3-Embed-0.6B$_{(ZS)}$	arXiv'25	Dense	0.3376	0.4852	0.5354	0.3973	0.3107
Qwen3-Embed-4B$_{(ZS)}$	arXiv'25	Dense	0.4410	0.5783	0.6013	0.4985	0.3793
Qwen3-Embed-0.6B$_{(FT)}$	arXiv'25	Dense	0.5771	0.6376	0.6818	0.6045	0.4460
Qwen3-Embed-0.6B + TempDate$_{(FT)}$	SIGIR'23	Dense	0.5523	0.6425	0.6630	0.5924	0.4391
Qwen3-Embed-0.6B + TempDate-Smooth$_{(FT)}$	PMLR'23	Dense	0.5638	0.6346	0.6727	0.5942	0.4396
CTD $_{\textbf{Qwen3-Embed-0.6B}}$ (Ours)	This work	Dense	**0.5923**	**0.6485**	**0.6927**	**0.6194**	**0.4575**

CTD beats lexical, time-aware sparse, and semantic dual-encoder baselines.

Analysis

Where the Gain Comes From

Table 3: Query-span analysis. CTD is consistently strong on both single-month and multi-month queries

Method	Single-month ($\|Q\| = 1$)		Multi-month ($\|Q\| > 1$)	
	R@1	MRR@10	R@1	MRR@10
BM25	0.397	0.413	0.396 ↓-0.001	0.499 ↑+0.086
ColBERT-LFM2 (ZS)	0.298	0.312	0.386 ↑+0.088	0.490 ↑+0.178
mE5-Large (ZS)	0.259	0.279	0.337 ↑+0.078	0.422 ↑+0.143
BERT-base (FT)	0.497	0.516	0.525 ↑+0.027	0.621 ↑+0.106
CTD BERT-base (Ours)	0.509	0.530	0.685 ↑+0.176	0.744 ↑+0.214
Qwen3-Embed-0.6B (ZS)	0.353	0.385	0.317 ↓-0.036	0.415 ↑+0.031
Qwen3-Embed-0.6B (FT)	0.481	0.495	0.711 ↑+0.230	0.757 ↑+0.262
CTD Qwen3-Embed-0.6B (Ours)	0.491	0.513	0.733 ↑+0.242	0.767 ↑+0.255

Takeaway: The improvement mainly comes from better temporal consistency, especially for window-style queries and chrono-near confounders. Multi-positive learning improves coverage, while temporal bias/context refine the top-ranked evidence.

Variant	$\mathcal{L}_{\text{multi}}$	b_{ij}^{time}	$\mathbf{c}_x$	R@1	MRR@10
FT baseline	–	–	–	0.5771 —	0.6044 —
+ $\mathcal{L}_{\text{multi}}$	✓	–	–	0.5820 ↑+0.0049	0.6107 ↑+0.0063
+ Bias	✓	✓	–	0.5898 ↑+0.0127	0.6135 ↑+0.0091
+ Ctx	✓	–	✓	0.5850 ↑+0.0079	0.6134 ↑+0.0090
Full (Ours)	✓	✓	✓	0.5923 ↑+0.0152	0.6194 ↑+0.0150

Table 4: Ablation on Qwen3-Embed.

Variant	$\mathcal{L}_{\text{multi}}$	b_{ij}^{time}	$\mathbf{c}_x$	R@1	MRR@10
FT baseline	–	–	–	0.5088 —	0.5597 —
+ $\mathcal{L}_{\text{multi}}$	✓	–	–	0.5178 ↑+0.0090	0.5685 ↑+0.0088
+ Bias	✓	✓	–	0.5384 ↑+0.0296	0.5776 ↑+0.0179
+ Ctx	✓	–	✓	0.5620 ↑+0.0532	0.5961 ↑+0.0364
Full (Ours)	✓	✓	✓	0.5826 ↑+0.0738	0.6193 ↑+0.0596

Table 12: Ablation on BERT-base.

Examples

Better Retrieval Leads to Better Historical Grounding

QUERY(point): 魯莊公四年三月期間，《春秋》主要記錄了什麼？
What does the Chunqiu mainly record in Duke Zhuang's Year 4, Month 3?

Baseline

#1 魯莊公三年三月。《春秋》及三傳皆不載此月之事。
#1 Duke Zhuang, Year 3, Month 3: no annals entry in this month.
#2 四年春，王二月，夫人姜氏饗齊侯于祝丘。三月，紀伯姬卒。
#2 In spring of Year 4, month 2, Lady Jiang feasted the Marquis of Qi at Zhuqiu. In month 3, Ji Boji died.
#3 四年春，王三月庚戌，盜殺蔡侯申。弒君賤者窮諸人…
#3 In spring of Year 4, month 3, gengxu: Cai Hou Shen was murdered...

Ours

#1 **★四年春，王三月，楚武王荊尸，授師孑焉，以伐隨…** ✓
#1 ★ In spring of Year 4, month 3, King Wu of Chu marched against Sui...
#2 魯莊公三年三月。《春秋》及三傳皆不載此月之事。
#2 Duke Zhuang, Year 3, Month 3: no annals entry in this month.
#3 魯桓公四年三月：經傳於是月史事闕如，無專條可錄。
#3 Duke Huan, Year 4, Month 3: no annals entry in this month.

QUERY(window): 在魯哀公七年十一月之後的一個月裡，《春秋》是否另有記事？
In the month after Duke Ai's Year 7, Month 11, is there any further entry in the Chunqiu?

Baseline

#1 魯哀公七年十一月。《春秋》經文及三傳於此月無事可書。
#1 Duke Ai, Year 7, Month 11: no annals entry in this month.
#2 **★魯哀公七年十二月。《春秋》及三傳皆不載此月之事。** ✓
#2 ★ Duke Ai, Year 7, Month 12: no annals entry in this month.
#3 魯哀公七年九月：經傳於是月史事闕如，無專條可錄。
#3 Duke Ai, Year 7, Month 9: no annals entry in this month.

Ours

#1 **★魯哀公七年十二月。《春秋》及三傳皆不載此月之事。** ✓
#1 ★ Duke Ai, Year 7, Month 12: no annals entry in this month.
#2 魯哀公七年十一月。《春秋》經文及三傳於此月無事可書。
#2 Duke Ai, Year 7, Month 11: no annals entry in this month.
#3 魯哀公七年二月。《春秋》及三傳皆不載此月之事。
#3 Duke Ai, Year 7, Month 2: no annals entry in this month.

QUERY(window): 魯昭公十七年四月之前半年之內，《春秋》主要記事是什麼？
What does the Chunqiu mainly record in the half-year before Duke Zhao's Year 17, Month 4?

Baseline

#1 <u>昭十七年傳稱黃帝以雲名官，故知縉雲黃帝時官名…</u>
#1 The commentary says that the Yellow Emperor named officials after clouds...
#2 <u>金天國號少皞，身號譙周云："金天氏能脩大皞之法…</u>
#2 The state title of Jintian was Shaohao; Qiao Zhou says...
#3 魯昭公十七年四月。《春秋》經文及三傳於此月無事可書。
#3 Duke Zhao, Year 17, Month 4: no annals entry in this month.

Ours

#1 **★魯昭公十七年二月。《春秋》及三傳皆不載此月之事。** ✓
#1 ★ Duke Zhao, Year 17, Month 2: no annals entry in this month.
#2 **★十有七年春，小邾子來朝。** ✓
#2 ★ In spring of Year 17, the ruler of Xiao Zhu came to court.
#3 魯昭公十七年十二月。《春秋》經文及三傳於此月無事可書。
#3 Duke Zhao, Year 17, Month 12: no annals entry in this month.

- CTD retrieves the right month, not just similar text.
- Better retrieval improves RAG grounding.
- Failure mode reduced: fluent but wrong-time answers become evidence-grounded historical answers.

System: 你是一名严谨的《春秋》史料问答助手。

Prompt:
你必须**只依据下方给定资料**回答，禁止使用外部常识补全。
如果资料不足以支持结论，请回答"资料不足"，并说明缺少什么。
【资料（可能包含同月注疏、相邻月近似事件、空月占位等干扰项）】
你将看到若干条DOC记录，请先判断哪些记录最能直接回答问题。

"凡卿出使，必具其名氏，以尊君命。今不書氏…"
"祭伯，諸侯為王卿士者。祭國，伯爵也。傳曰："非王命也。"…"
…

Ground truth DOCs

"冬，十有二月，祭伯來。祭伯者何？天子之大夫也。何以不稱使？奔也。…"
"冬，十有二月，祭伯來。來者，來朝也，其弗謂朝，何也？寰內諸侯…"
"公子益師卒。"
"公子益師卒。何以不日？遠也。所見異辭，所聞異辭，所傳聞異辭。"
"公子益師卒。大夫日卒，正也；不日卒，惡也。"

…
"二年春，公會戎于潛。"
"二年春，公會戎于潛。會者，外為主焉爾。知者慮，義者行，仁者守。…"
…

【问题】
查詢《春秋》，魯隱公元年十二月這一月有什麼重要事件？
【你必须按以下步骤输出】
1) 证据选择：从 DOC 中选出"最关键"的 1~3 条。
2) 最终回答：只基于你选的证据作答（中文，2~6 句）。
3) 证据引用：在答案末尾列出你用到的 DOC 。
4) 若问题对应月份为空月：请明确写"该月无事可书"（以资料为准）。

Output (w/o Thinking):

Ground Truth coverage 2/2 (correct)

1) 证据选择：
（魯隱公元年十二月）：这是问题直接对应的月份记录，包含两条经文及三传相关记载，是回答问题的核心依据。

2) 最终回答：
根据《春秋》记载，鲁隐公元年十二月发生了两件重要事件。<u>第一，祭伯来到鲁国</u>，但《左传》指出这是"非王命也"，即并非奉周天子之命。<u>第二，鲁国公子益师去世</u>，但《春秋》未记载其去世的具体日期。因此，该月的重要事件是祭伯来和公子益师卒。

3) 证据引用：
Evidence:

Grounded: cites ground truth DOCs

"冬，十有二月，祭伯來。祭伯者何？天子之大夫也。何以不稱使？奔也。…"
"公子益師卒。"

Ground Truth: (1) 祭伯来；(2) 公子益师卒。

Conclusion

Time Matters at Retrieval Time

1. **New Task**
   Time-keyed temporal retrieval for non-Gregorian classical Chinese annals.

2. **New Benchmark**
   ChunQiuTR with reign-based month keys, no-event months, and chrono-near confounders.

3. **New Method**
   CTD: calendrical dual-encoder with absolute context, relative bias, and interval-overlap supervision.

Query → Time-keyed retrieval → Evidence-grounded answer

Right evidence = relevant + temporally faithful.

THANK YOU!

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Code & Data: **github.com/xbdxwyh/ChunQiuTR**